

Complete School ERP System Manual

Professional Overview & Workflow Guide for Schools

This document presents a complete School ERP (Enterprise Resource Planning) system designed to manage school operations digitally. The ERP helps automate administration, student management, attendance, fees, academics, staff operations, examinations, communication, and reporting.

1. Dashboard Module

- Admin dashboard with overall school statistics
- Teacher dashboard with classes and attendance
- Student dashboard with results and fees
- Role-based access control

2. Student Management

- Add/Edit/Delete student records
- Student profile with photo and personal details
- Parent and guardian management
- Session-wise student records

3. Attendance Management

- Daily attendance tracking
- Class-wise attendance
- Teacher attendance
- Attendance reports and analytics

4. Fee Management

- Fee structure setup
- Fee collection and receipts
- Pending fee reports
- Online payment support

5. Examination & Result

- Exam scheduling
- Marks entry system
- Result generation
- Progress reports

6. Session Management

- Multiple academic sessions
- Single active editable session
- Read-only inactive sessions
- Student promotion system

7. Staff & Teacher Management

- Teacher profiles
- Department management
- Salary tracking
- Leave management

8. Timetable Module

- Class routine generation
- Teacher period allocation
- Subject scheduling

9. Communication System

- SMS notifications
- Parent communication
- Announcements and notices

10. Reports & Analytics

- Student performance reports
- Attendance reports
- Financial reports
- Export to PDF and Excel

ERP Workflow Overview

The School ERP follows a modular workflow where administrators manage admissions, teachers manage attendance and academic activities, accountants handle fees, and students/parents access information through dashboards.

User Role	Main Responsibilities
Admin	Complete ERP management and monitoring
Teacher	Attendance, marks, class activities
Accountant	Fee collection and financial reports
Student	View attendance, marks, notices
Parent	Track child performance and fees

Technology Stack

Frontend: Vite + Tailwind CSS + JavaScript

Backend: Node.js + Express.js

Database: MongoDB / PostgreSQL

Authentication: JWT-based secure login system

Deployment: Cloud/VPS hosting support

Security Features

- Secure JWT authentication
- Role-based access control
- Protected admin routes
- Session-based data isolation
- Backup and recovery support

Benefits for Schools

- Reduces paperwork and manual work
- Improves communication with parents
- Centralized digital management
- Fast report generation
- Better academic monitoring
- Improved transparency and efficiency

This ERP system is scalable and suitable for schools, colleges, and educational institutions. It can be customized according to institution requirements.